



DEPARTMENT OF SOFTWARE ENGINEERING
Semester-V

LAB 1 — SOLUTION REPORT

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Lab Number: 01

Lab Title: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Areeba Habib

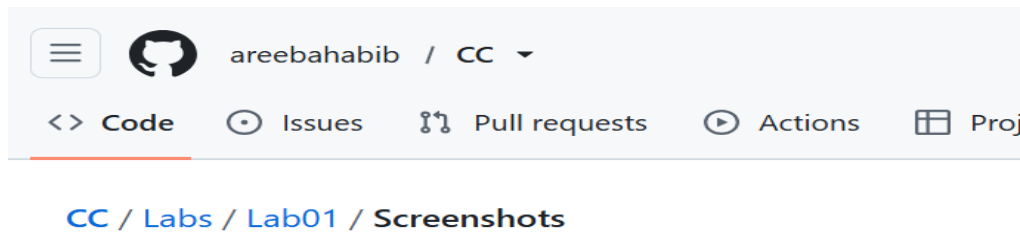
Registration Number: FA24B1-SE-011

Course Code: BSE-665

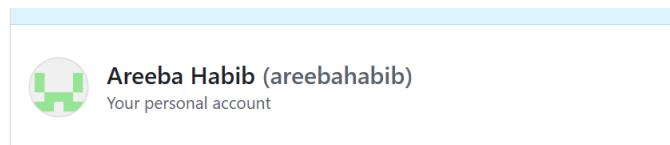
Section: A

GitHub Username: areebahabib

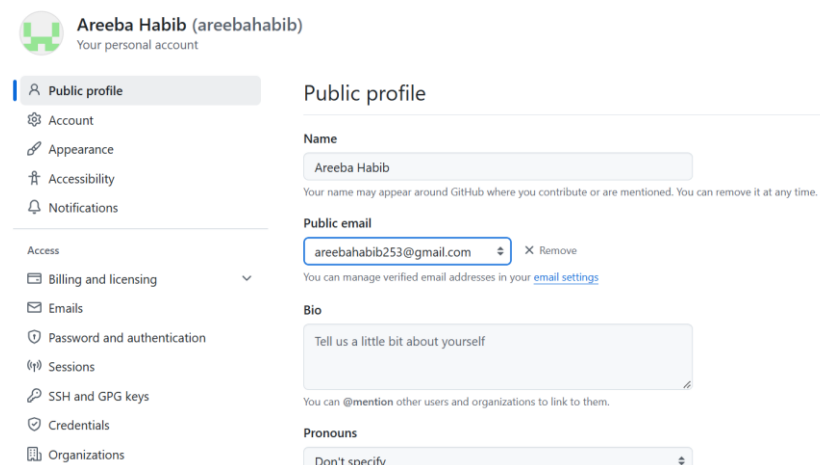
Getting Started



Task 1: Create and Configure Your GitHub Account

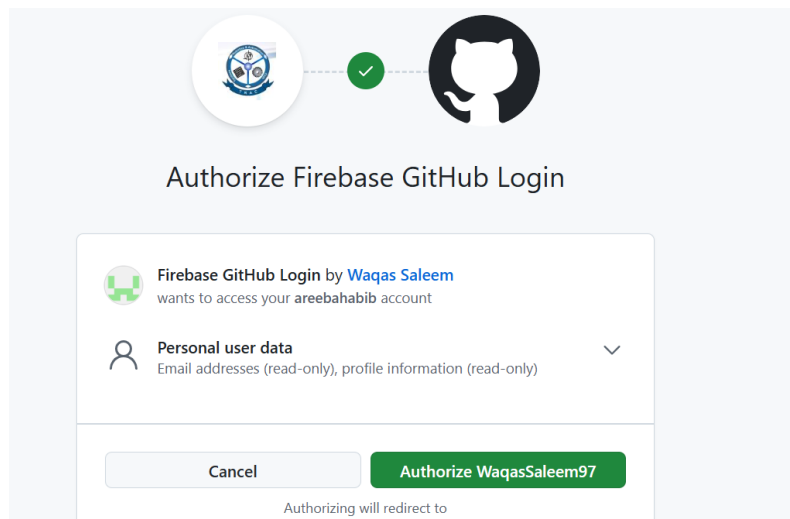


GitHub account header confirming the personal account “areebahabib” has been created successfully.



GitHub “Public profile” settings page showing the account's Name field updated to the student's actual name, Areeba Habib.

Task 2: Register in the Student Marks Portal



Student Marks Portal registration form, showing the registration number, course (SE-665 – Cloud Computing), and section (Section A) being entered.

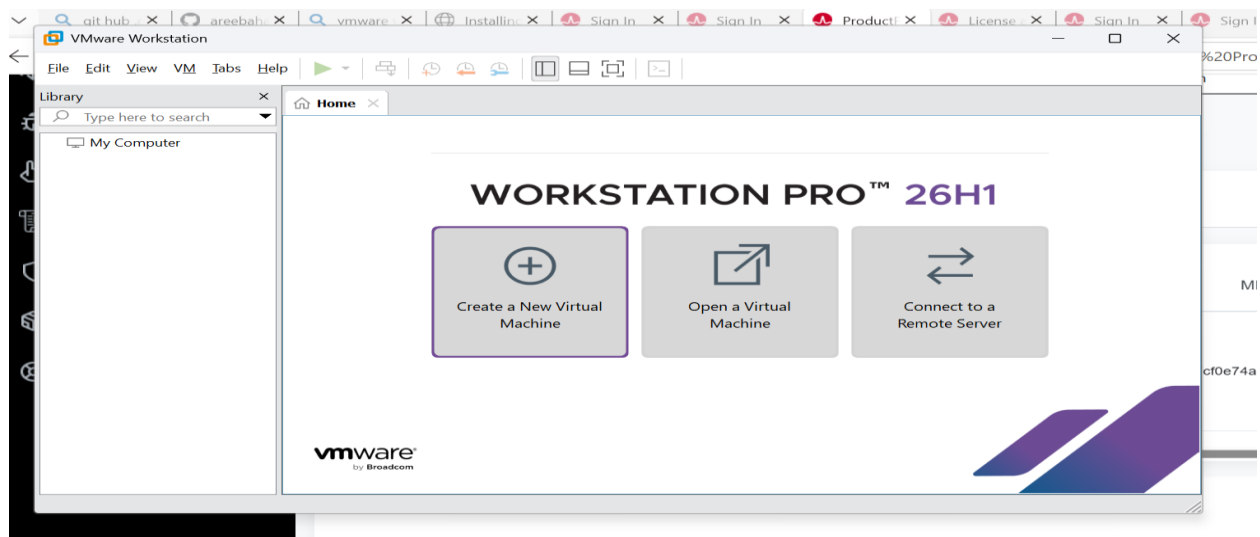
A screenshot of the "Complete registration" form. At the top, there is a black circle with the white text "ID". Below this is the heading "Complete registration" in bold. Underneath, it says "Enter your registration number, course and section. The enrollment requires approval." The form has three input fields: "Registration number" with the value "2024-BSE-11", "Course" with the value "SE-665 — Cloud Computing", and "Section" with the value "Section A". Below these fields is a dark blue button labeled "Submit enrollment" and a light blue button labeled "Sign out". A small note below the registration number field says "Allowed: 2024-BSE-01 to 2024-BSE-99."

GitHub authorization prompt for the Student Marks Portal, confirming that the portal signs in using the student's GitHub identity.



Student Marks Portal dashboard showing the Cloud Computing (SE-665), Section A enrollment submitted and awaiting administrator approval.

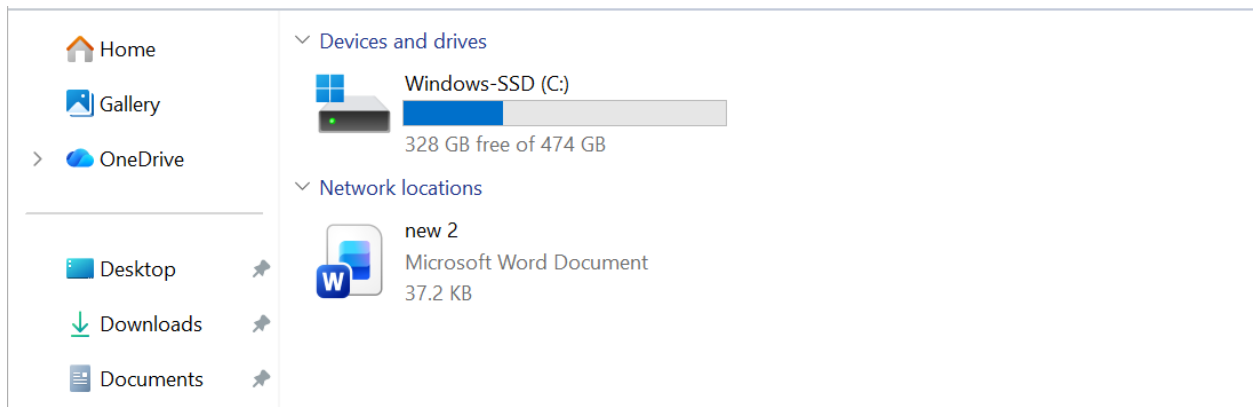
Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro launch screen, confirming successful installation of the virtualization software.

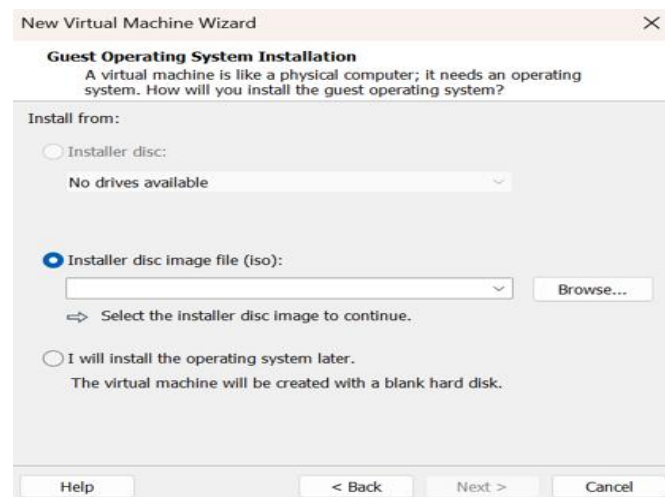


Downloaded Ubuntu Server installer file (.iso) shown in File Explorer.

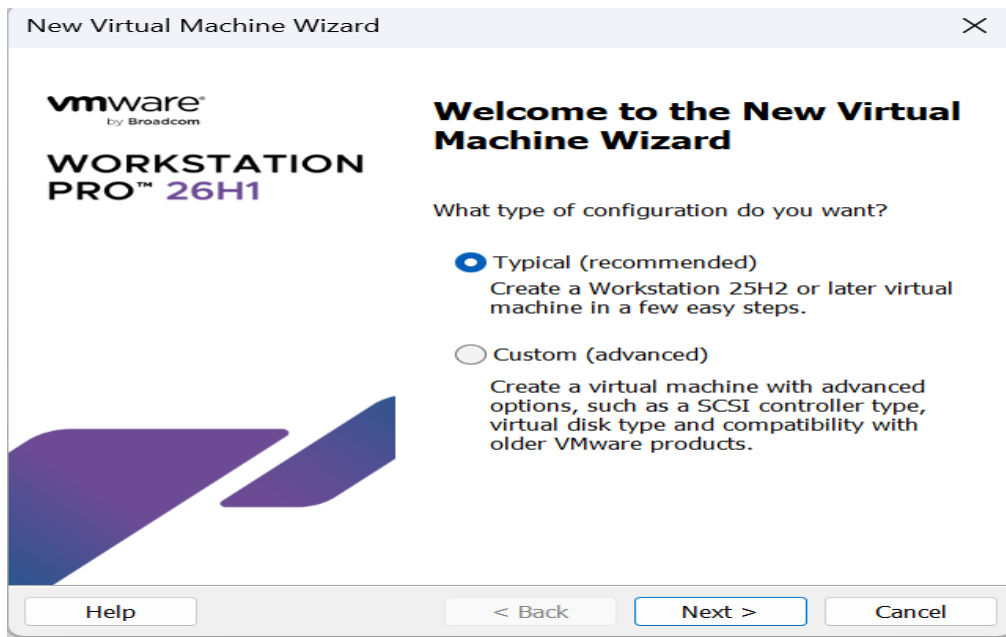


Windows storage panel confirming sufficient free disk space is available to create and run the virtual machine.

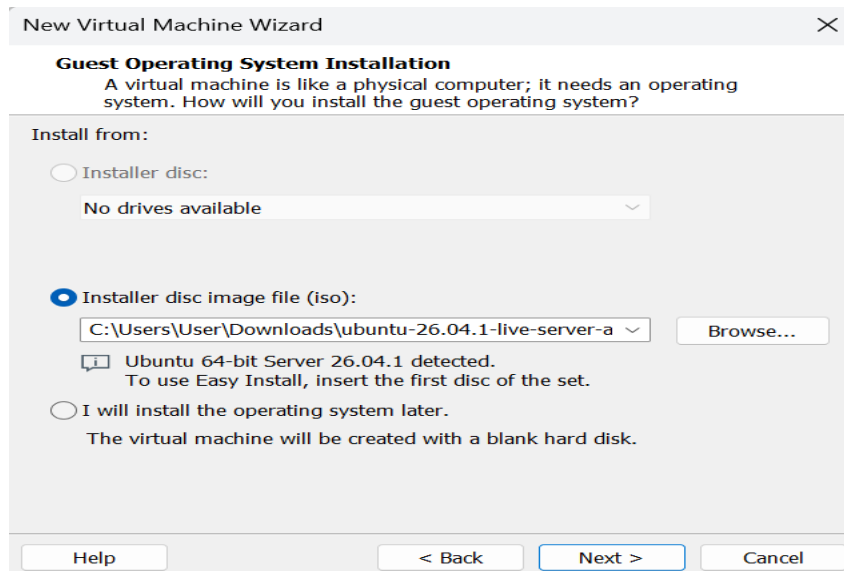
Task 4: Create the Ubuntu Server Virtual Machine



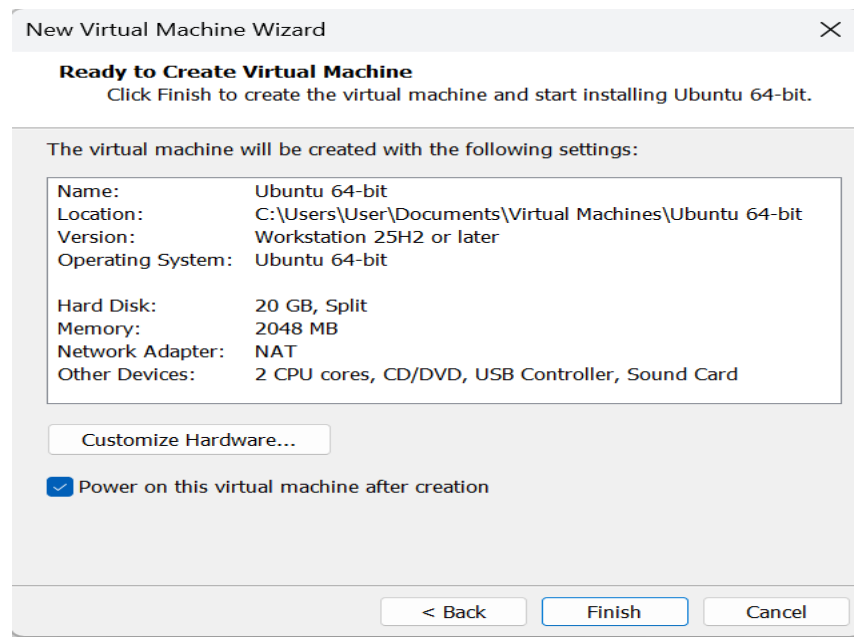
New Virtual Machine Wizard — selecting the Ubuntu Server ISO as the installer disc image for the guest operating system.



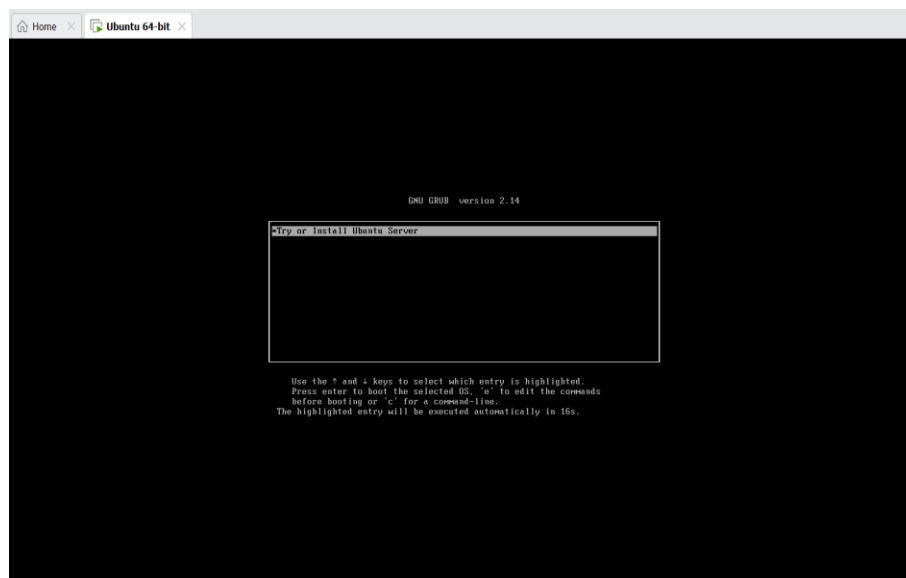
New Virtual Machine Wizard welcome screen with the **“Typical (recommended)”** configuration option selected.



New Virtual Machine Wizard automatically detecting the Ubuntu 64-bit Server ISO for Easy Install.

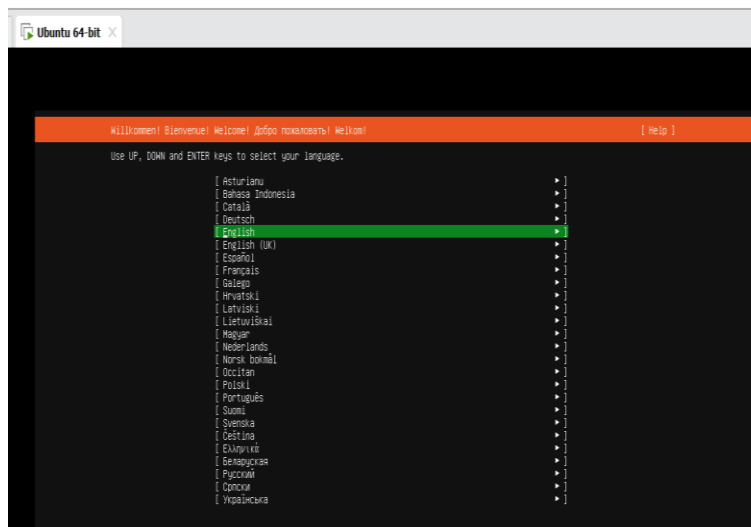


Summary screen listing the finalized VM settings (disk size, memory, networking) before creation.

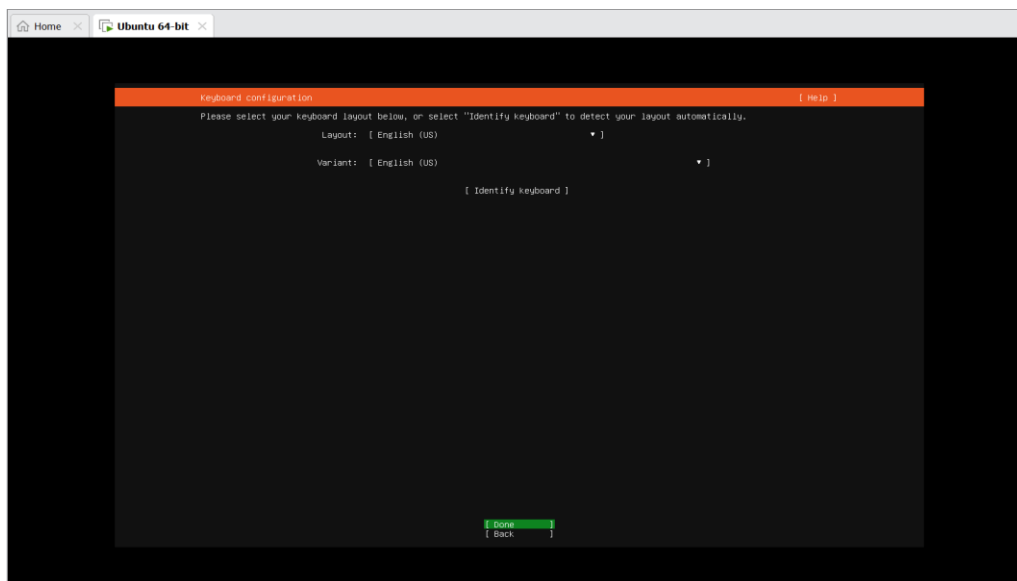


Initial boot screen of the Ubuntu Server installer inside the newly created virtual machine.

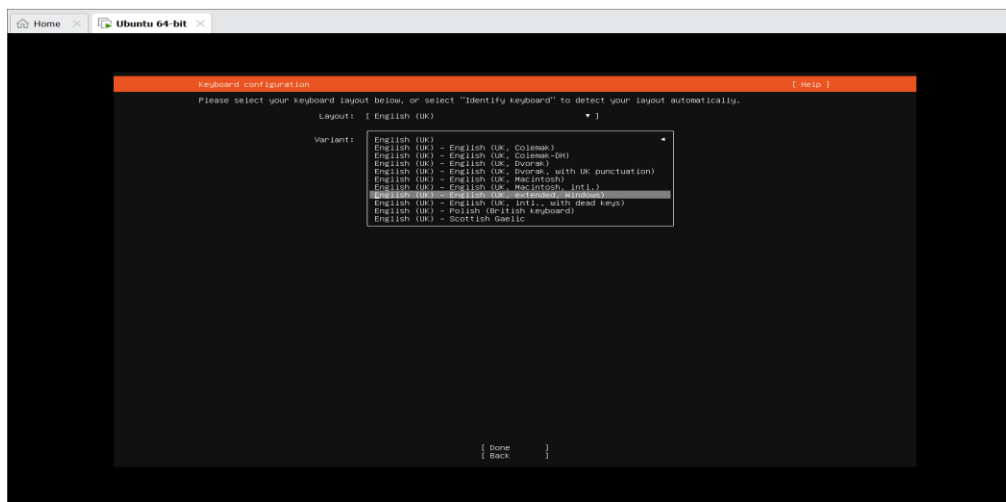
Task 5: Choose the Installation Language



Ubuntu installer language selection screen with **“English”** highlighted as the chosen installation language.

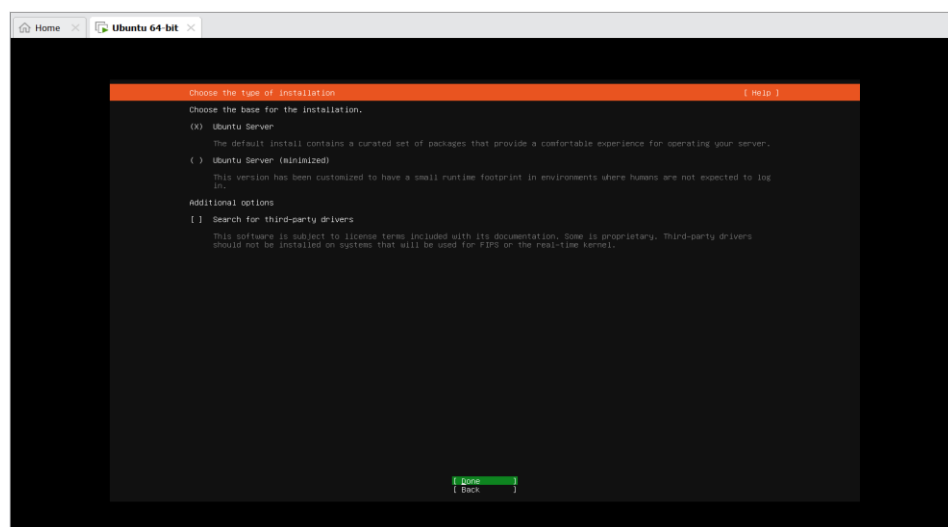


Keyboard configuration screen for selecting the keyboard layout during installation.



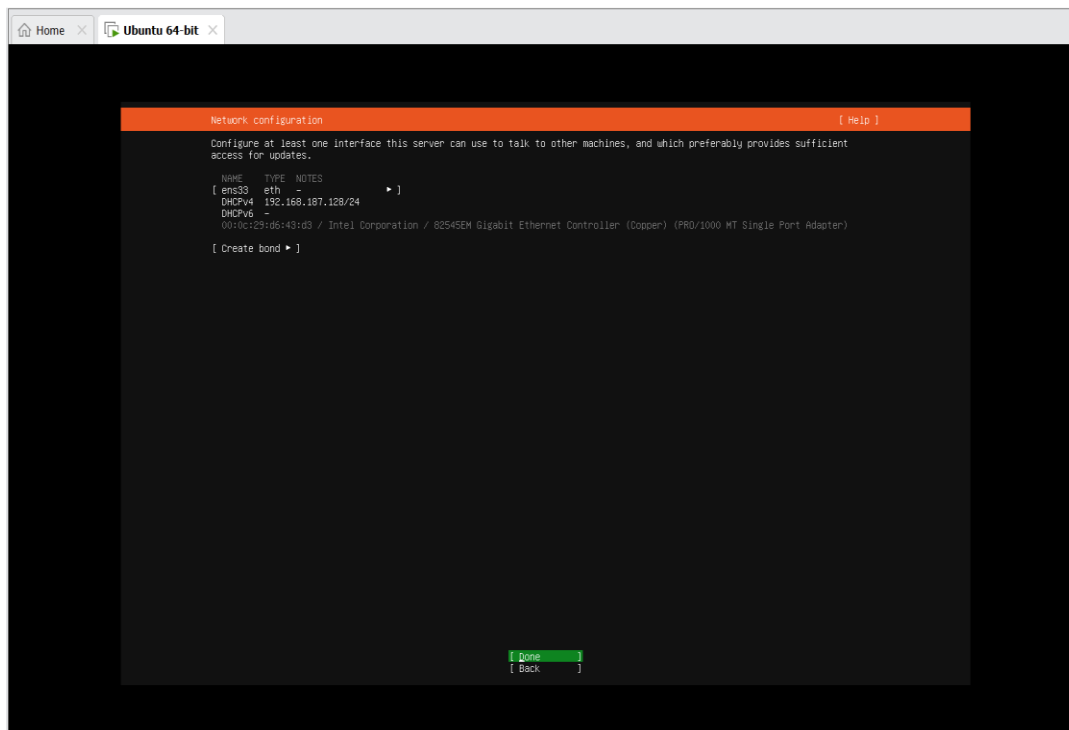
Keyboard configuration screen with the list of keyboard layout variants expanded for selection.

Task 7: Choose the Ubuntu Server Installation Type

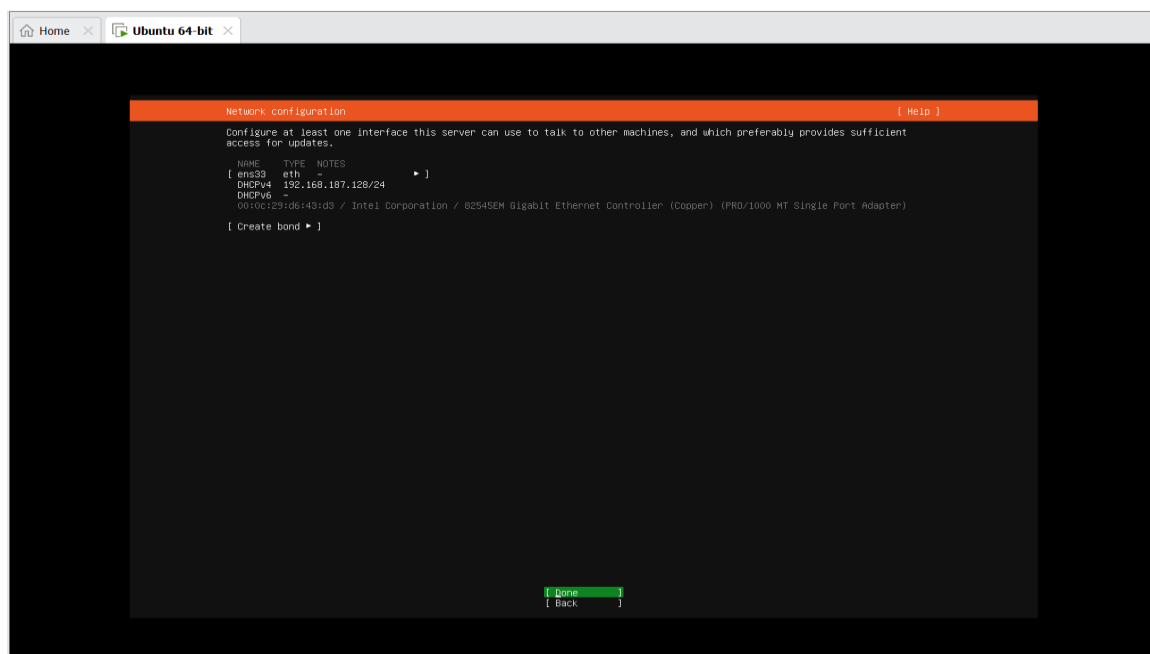


Installation type screen with “**Ubuntu Server**” selected as the base system to install.

Task 8: Configure the Network

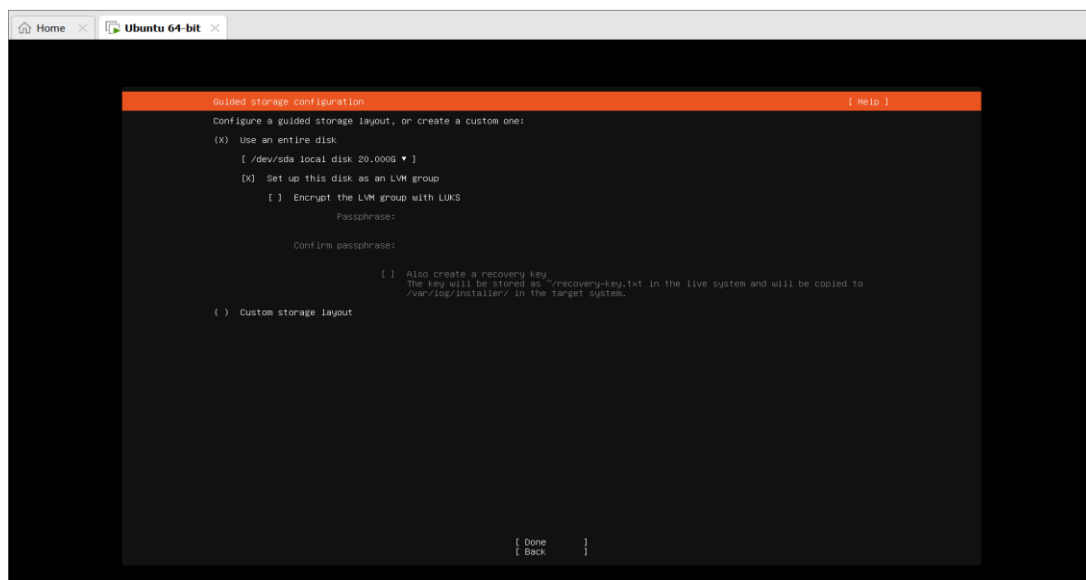


Network configuration screen showing the network interface automatically assigned an IP address via DHCP.

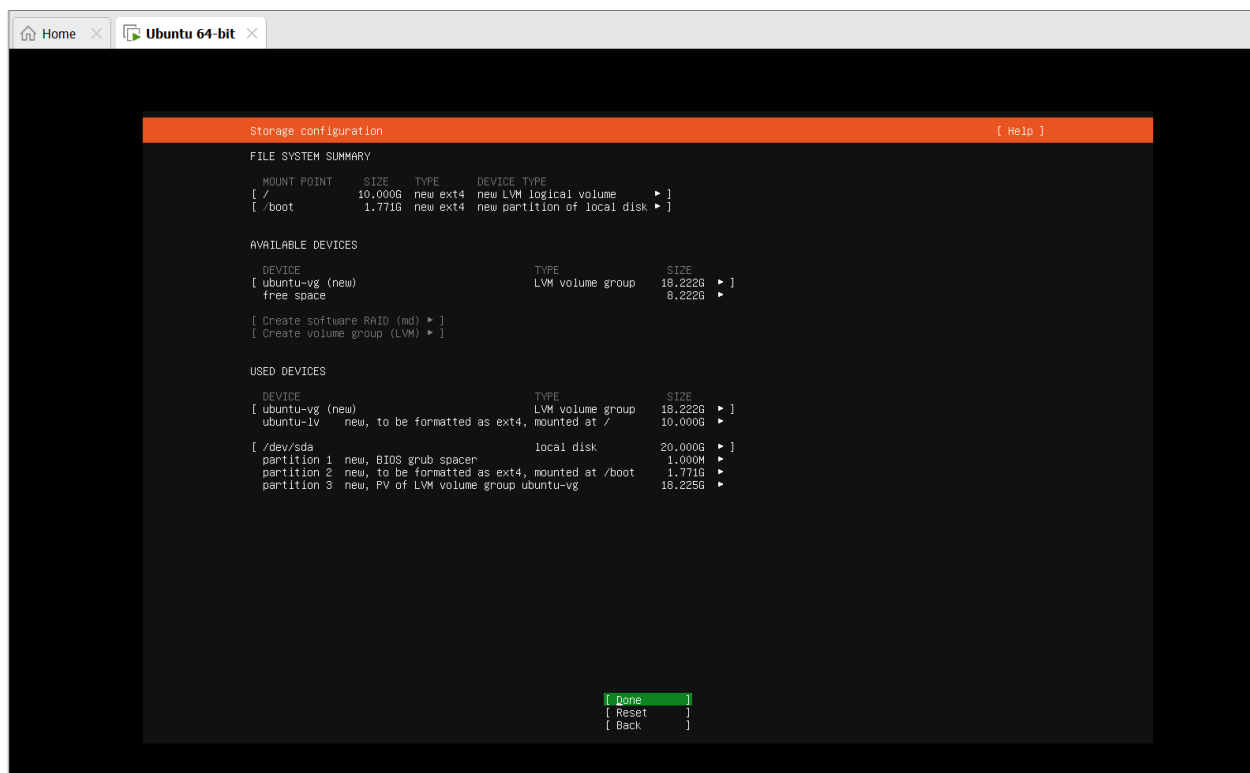


Network configuration confirmed, allowing the installer to proceed with the detected network interface.

Task 9: Configure Storage



Guided storage configuration screen with **“Use an entire disk”** selected for the installation.



File system summary showing the LVM volume group created on top of the selected disk.

Task 10: Select the Installation Disk

```

Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G  new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶

```

Storage configuration summary confirming the target disk selected for the Ubuntu Server installation.

```

Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G  new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
free space      8.222G ▶

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

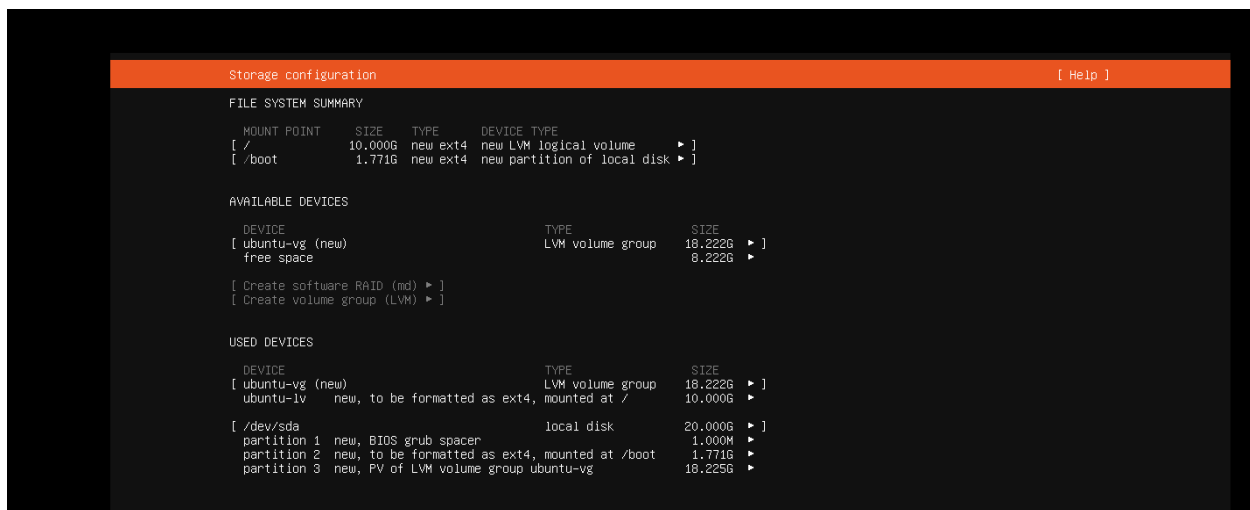
DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 18.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶

[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶
partition 3     new, PV of LVM volume group ubuntu-vg 18.225G ▶

```

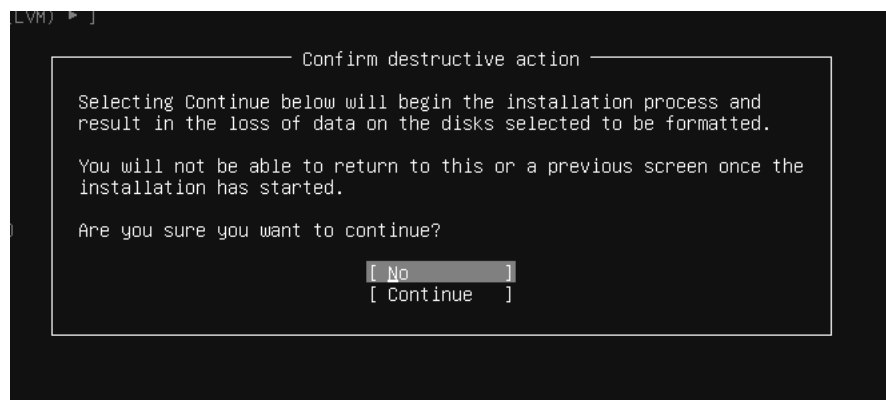
Detailed storage layout listing the available and used devices, including the LVM volume group and partitions.

Task 11: Review the Partitions

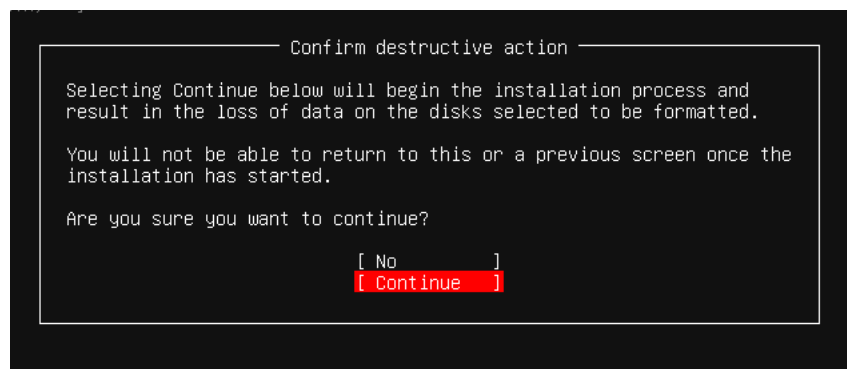


Final review of the partition layout — boot, root, and swap partitions — before applying the storage changes.

Task 12: Confirm the Storage Changes

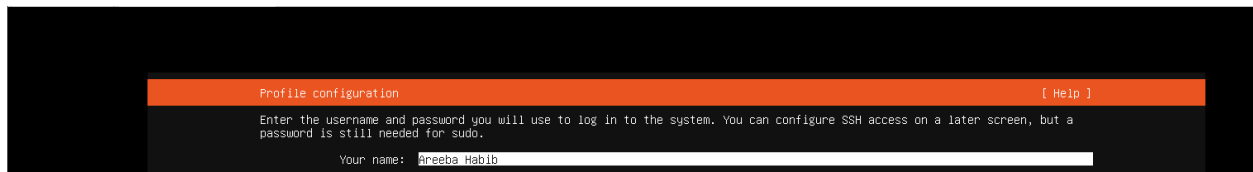


“Confirm destructive action” warning dialog, alerting that continuing will erase and format the selected disk.

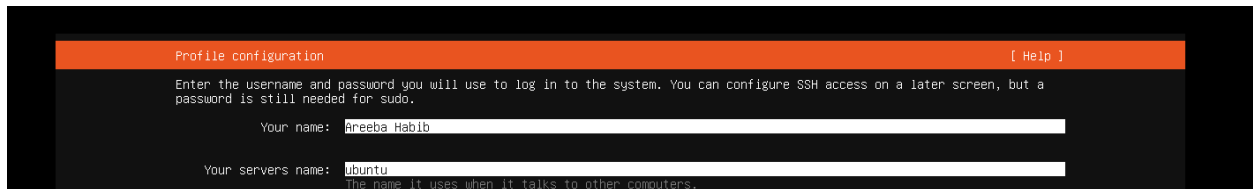


Confirmation dialog with “Continue” selected, proceeding to apply the storage changes and begin installation.

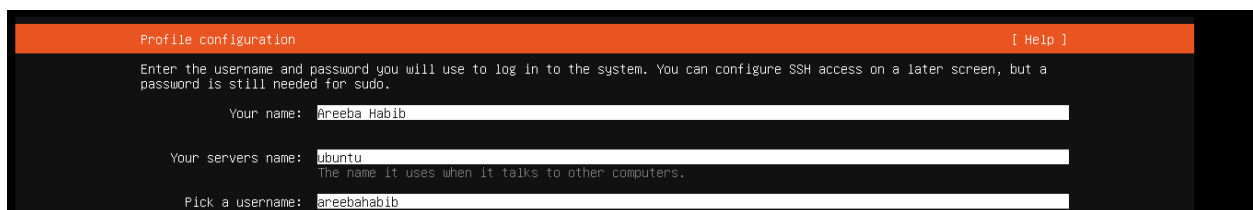
Task 13: Configure the Ubuntu Profile



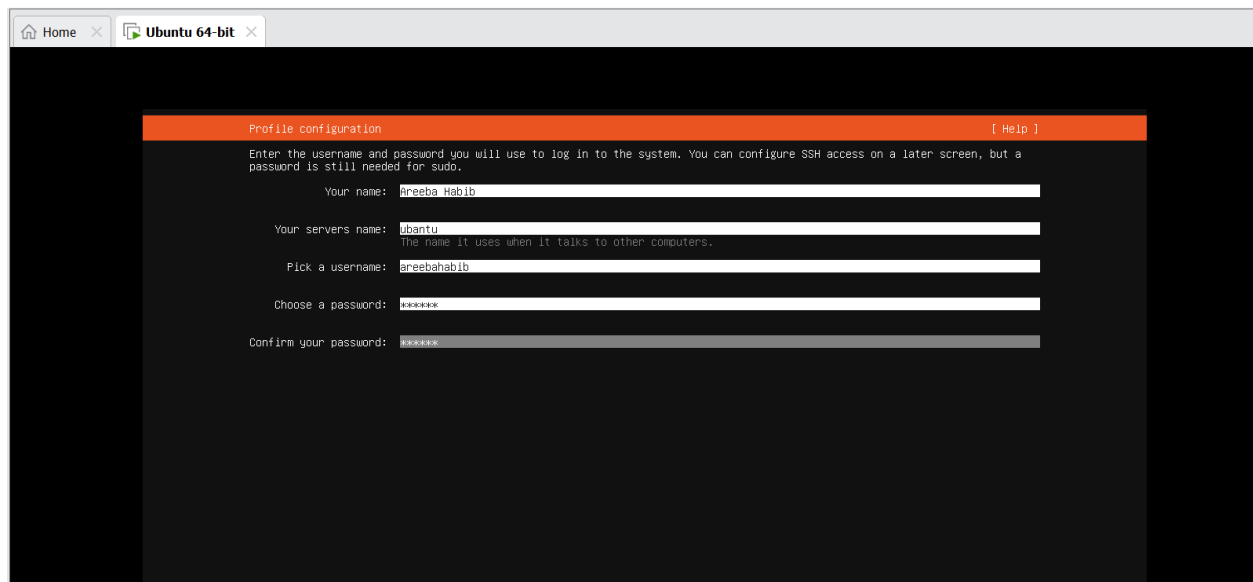
Profile configuration screen — entering the account holder's real name, Areeba Habib.



Profile configuration screen — setting the server's hostname.



Profile configuration screen — setting the login username, areebahabib.



Completed profile configuration screen with name, server name, username, and password all filled in.

Task 14: Complete the Software Installation

```

Installing system
subiquity/2dev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
configuring storage
  running 'curtin block-meta simple'
  curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring partition: partition-2
  configuring format: format-0
  configuring lvm_voigroup: lvm_voigroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
  acquiring and extracting image from cp:///tmp/tmpxbk4_i7l/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
  configuring apt configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel \

```

Installer progress log showing the base system, kernel, and packages being installed onto the virtual disk.

Task 15: Reboot and Verify the Ubuntu Identity

```

Installation complete!
  running 'curtin extract'
  curtin command extract
  acquiring and extracting image from cp:///tmp/tmpxbk4_i7l/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
  configuring apt configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
  calculating extra packages to install
  installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
  configuring cloud-init
  downloading and installing security updates
  curtin command in-target
  restoring apt configuration
  curtin command in-target
subiquity/Late/run_builtins:
subiquity/Late/run_user_supplied:

```

View full log
[Reboot Now]

“Installation complete!” screen with the option to reboot the newly installed Ubuntu Server.

```
ubuntu login:
ubuntu login: areebahabib
Password:
```

Caption: Ubuntu login prompt appearing after the virtual machine reboots into the freshly installed system.

```
ubuntu login: areebahabib
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 02:16:23 PM UTC 2026

System load:  0.02          Processes:           218
Usage of /:   50.8% of 9.75GB Users logged in:          0
Memory usage: 22%          IPv4 address for ens33: 192.168.187.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

areebahabib@ubuntu:~$ _
```

Successful login banner for Ubuntu Server, displaying system information after authenticating as areebahabib.

```
areebahabib@ubuntu:~$ _
```

Terminal prompt “areebahabib@ubuntu:~\$” confirming the configured username and hostname match the profile set up earlier.

Task 16: Find the Ubuntu Server IP Address

```
areebahabib@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:f3:f2:b5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29f3f2b5
    inet 192.168.187.132/24 metric 100 brd 192.168.187.255 scope global dynamic ens33
        valid_lft 1605sec preferred_lft 1605sec
    inet6 fe80::20c:29ff:fef3:f2b5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
areebahabib@ubuntu:~$ _
```

Output of the `ip addr` command listing the server's network interfaces.

```
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:f3:f2:b5 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29f3f2b5
    inet 192.168.187.132/24 metric 100 brd 192.168.187.255 scope global dynamic ens33
        valid_lft 1605sec preferred_lft 1605sec
    inet6 fe80::20c:29ff:fef3:f2b5/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
areebahabib@ubuntu:~$ _
```

Close-up of the network interface entry, showing the server's assigned IPv4 address.

Task 17: Connect to Ubuntu Server from Windows

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh areebahabib@192.168.187.132
The authenticity of host '192.168.187.132 (192.168.187.132)' can't be established.
ED25519 key fingerprint is SHA256:WUoB4Y0HayD2NZTCF7eQDThQGSJUetj61PnVX0QQH9E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Windows Command Prompt running the `ssh` command to connect to the server.

```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.187.132' (ED25519) to the list of known hosts.
areebahabib@192.168.187.132's password: |
```

SSH client prompting to confirm the server's host key fingerprint before connecting for the first time.

```
areebahabib@192.168.187.132's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 02:42:58 PM UTC 2026

System load:  0.0               Processes:            222
Usage of /:   51.0% of 9.75GB   Users logged in:     0
Memory usage: 25%              IPv4 address for ens33: 192.168.187.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

areebahabib@ubuntu:~$ |
```

Successful SSH login banner showing Ubuntu Server system information after connecting remotely from Windows.

```
areebahabib@ubuntu:~$ |
```

Terminal prompt **"areebahabib@ubuntu:~\$"** confirming an active remote SSH session from the Windows host.